

Chromatin Signature and Transcription Factor Binding Provide a Predictive Basis for Understanding Plant Gene Expression

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Chromatin accessibility and post-transcriptional histone modifications play important roles in gene expression regulation. However, little is known about the joint effect of multiple chromatin modifications on the gene expression level in plants, despite that the regulatory roles of individual histone marks such as H3K4me3 in gene expression have been well-documented. By using machine-learning methods, we systematically performed gene expression level prediction based on multiple chromatin modifications data in *Arabidopsis* and rice. We found that as few as four histone modifications were sufficient to yield good prediction performance, and H3K4me3 and H3K36me3 being the top two predictors with known functions related to transcriptional initiation and elongation, respectively. We demonstrated that the predictive powers differed between protein-coding and non-coding genes as well as between CpG-enriched and CpG-depleted genes. We also showed that the predictive model trained in one tissue or species could be applied to another tissue or species, suggesting shared underlying mechanisms. More interestingly, the gene expression levels of conserved orthologs are easier to predict than the species-specific genes. In addition, chromatin state of distal enhancers was moderately correlated to gene expression but was dispensable if given the chromatin features of the proximal regions of genes. We further extended the analysis to transcription factor (TF) binding data. Strikingly, the combinatorial effects of only a few TFs were roughly fit to gene expression levels in *Arabidopsis*. Overall, by using quantitative modeling, we provide a comprehensive and unbiased perspective on the epigenetic and TF-mediated regulation of gene expression in plants.

Keywords: Chromatin modification • Enhancer • Gene expression • Machine learning • Plant • Transcription factor.

Introduction

In eukaryotes, DNA is packaged in the form of chromatin. The basic unit of chromatin is the nucleosome, which consists of 147 bp of DNA wrapped around a histone octamer comprising

two H2A-H2B dimers and one H3-H4 tetramer. These histone proteins have tails where many residues can be post-translationally modified to influence chromatin compaction, nucleosome dynamics and gene transcriptions (Lawrence et al. 2016). Recently, the effects of histone modifications on gene activation and silencing have increasingly been considered important in animals. For example, active genes typically carry high levels of lysine methylation on the H3 and H4 tails, including trimethylation of H3 lysine 4 (H3K4me3) (Santos-Rosa et al. 2002), lysine 79 (H3K79me3) (Dong and Weng 2013) and lysine 36 (H3K36me3) (Chantalat et al. 2011). Some other histone modifications, such as tri-methylation of H3 lysine 9 (H3K9me3) and lysine 27 (H3K27me3), are associated with transcriptional suppression (Zhang et al. 2015). Therefore, transcriptionally active and silent chromatin is generally characterized by distinct modifications on the histone or combinations thereof (Zhang et al. 2015). Deciphering the 'histone code' for gene expression regulation has become an important research area (Dong and Weng 2013, Lawrence et al. 2016).

Extensive studies have demonstrated the associations between gene expression and histone modifications in animals, but fewer studies of this type have been performed using plants. Plants are sessile organisms, so performing rapid responses to external signals by activating signaling pathways that orchestrate the activities of responding genes is especially important for their survival (Van Dijk et al. 2010). Changes in histone modifications have been shown to be correlated with altered gene expression in plants under various stresses (Sokol et al. 2007, Zong et al. 2013, Fortes and Gallusci 2017). And some types of histone modifications have been investigated in the context of gene expression regulation in plants (Pfluger and Wagner 2007, Brusslan et al. 2015). For example, among H3K4me1, H3K4me2 and H3K4me3, only H3K4me3-containing genes were highly expressed in the *Arabidopsis* seedling tissue (Zhang et al. 2009). In addition, H3K9ac was also found to be positively correlated with the gene expression levels in *Arabidopsis*; however, H3K27me3 was generally enriched in lowly expressed genes (Ha et al. 2011). Although these pioneering studies have made progress in establishing the effects of individual chromatin modifications on plant gene expression, there is no comprehensive study on the quantitative

relationship between the combinatorial effects of multiple chromatin modifications and gene expression levels in plants.

Recently, machine-learning-based methods have been applied to fit models to data regarding histone modifications and gene expression in animals. For example, support vector regression (SVR) model was used to probe the relations between gene expression levels and the intensity of histone modifications; consequently, a strong correlation was revealed in worms (Cheng et al. 2011). A similar analysis in mouse embryonic stem cells was conducted by adding transcription factor (TF) binding information (Cheng and Gerstein 2012). Karlic et al. (2010) developed qualitative models to predict gene expression using a combination of 38 histone modifications in human CD4+ T-cells. Because gene expression levels can be experimentally measured by different techniques, Dong et al. (2012) built a novel quantitative model to explore the relationship between gene expression and histone modifications in the context of RNA sequencing. More recently, a deep learning-based method was conducted and demonstrated a state-of-the-art performance on gene expression prediction from histone modifications (Singh et al. 2016). Thus, these studies have revealed the relationships between gene expression and histone modifications in unprecedented detail, providing insight into the relative biological significance of individual changes in gene expression in animals.

In contrast, the quantitative relationship between chromatin modifications and gene expression in plants merits further investigation. Some important issues related to this area of investigation are as follows: (i) Can histone modifications be used to accurately predict gene expression levels in plants? (ii) What is the minimal combination of histone modifications that offer sufficient information about plant gene expression? (iii) Whether the quantitative relationship between gene expression and histone modifications is conserved across plant tissues and species? (iv) How is the regulatory information encoded in enhancers adopted by plant cells toward gene expression? and (v) Can TF-binding information be predictive to plant gene expression levels?

Besides the histone modifications, genomic DNase I hypersensitive sites (DHSs), which canonically mark open chromatin regions in genomes have been widely profiled to map the accessible chromatin landscape in higher animals (Thurman et al. 2012) and they are also associated with gene expression patterns (Natarajan et al. 2012). To answer the questions outlined above, using *Arabidopsis* and rice as two representative species of angiosperms, we integrated datasets for nine histone modifications and one DHS (hereinafter referred to as 'chromatin modifications') from *Arabidopsis* leaves, and datasets for 10 histone modifications and one DHS from rice leaves, respectively. We quantitatively analyzed their relationships with gene expression levels using machine-learning-based models. Our results indicate that the combinatorial signal of chromatin modifications in the proximal region of genes is a good predictor of gene expression levels. We also revealed that models trained using chromatin data from leaf tissue can well predict the gene expression levels in root tissue and vice versa. To understand how the regulatory signal is conveyed from

proximal to distal *cis*-regulatory elements toward gene expression level, we compared the predictive performance between models with or without using the chromatin features of enhancers. Additionally, we discovered that a few TFs could predict gene expression levels based on their ChIP-Seq profiles. Finally, our cross-species comparative analysis displayed that the predictive models have good generalization capacity, providing evidence that the relationship between chromatin modifications and gene expression is evolutionarily conserved between *Arabidopsis* and rice. Overall, we systematically analyzed the relations of plant gene expression levels with the chromatin features of *cis*-regulatory elements and the binding properties of *trans*-regulatory factors by using quantitative modeling.

Results

Gene expression levels in *Arabidopsis* can be predicted using chromatin modifications

A prerequisite for quantitatively assessing the relationship between gene expression and histone modifications is the accurate measurement of the histone modification levels of genes. To provide a basis for the assignment of modification levels to associated genes, we computed the relative signal intensity of each chromatin modification along the gene body and flanking region of each gene (Supplementary Fig. S1). Obviously, the signals of all inspected histone modifications were mainly present in gene body regions while the DHS signal prominently appeared at the immediately upstream of transcription start site (TSS). So, the combinatorial signals derived from the histone modifications within gene body regions and the DHS signal within 1 kb upstream of TSS (hereinafter called 'Type aware') were first used to model gene expression level. Considering that linear regression, random forest and SVR are widely used statistical models, each has its own characteristics and advantage in regression analysis, we trained the three selected models separately by using a particular dataset including the gene expression data and the chromatin modification data from a variety of sources (Supplementary Table S1). We performed 10-fold cross-validation to evaluate the model performance. The observed gene expression levels were highly correlated with the corresponding values predicted by the models (Fig. 1A–C). The two nonlinear models performed only slightly better than the linear regression model (linear regression: $r = 0.80$; random forest: $r = 0.84$; SVR: $r = 0.84$). Finally, we did not find a correlation between the predicted and measured expression levels by permutation analysis (Supplementary Fig. S2). Therefore, the observed high correlation between gene expression levels and the levels of chromatin modifications in *Arabidopsis* suggests a strong functional link between these processes.

Next, we determined the importance of features in the prediction of gene expression using linear regression and random forest models. H3K4me3, H3K36me3, H3K9ac and H3K27ac were the top four predictors in both linear regression and random forest models, whereas H3K27me1 and H3K27me3 were less important (Fig. 1D, E). This pattern is consistent

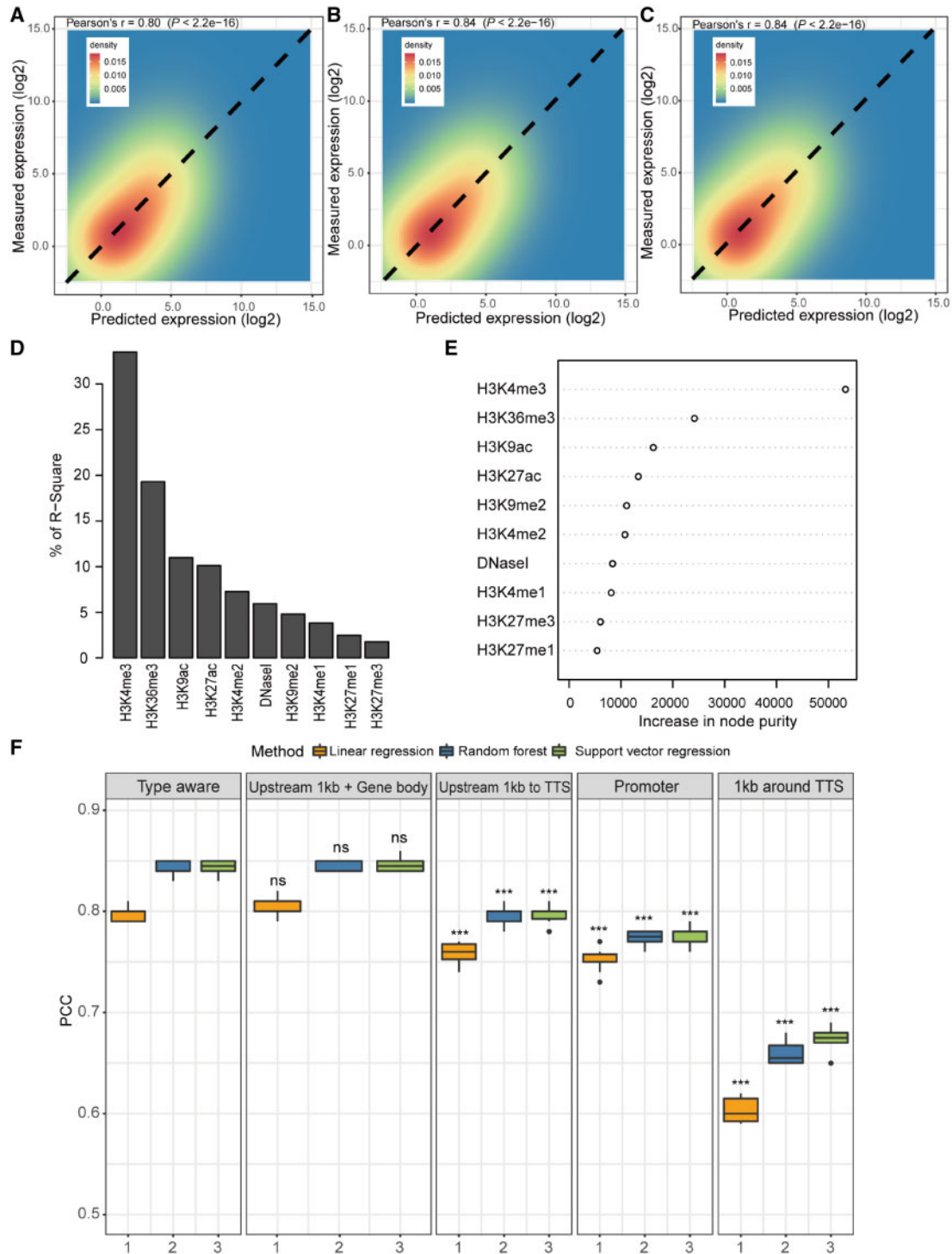


Fig. 1 Prediction of gene expression levels in *Arabidopsis* leaf tissue based on chromatin modifications. (A–C) 2D kernel density plots showing the predictive power of the linear regression, random forest and SVR models, respectively. The black line in each graph indicates the linear fit between the predicted and measured expression levels. (D, E) The feature importance of the linear regression model and random forest model, respectively. (F) The predictive power of models using chromatin modification signals measured in different genomic regions or in different ways. For the ‘Type aware’ model, the histone features were calculated using the signal within gene body, while the DHS feature values were derived from the regions 1 kb upstream to TSS. For the ‘Upstream 1 kb + Gene body’ model, its feature variables can be categorized into two parts: the first half contained the signal values of all chromatin modifications derived from the regions upstream 1 kb of TSS, while those of the second half were derived from the gene body regions. The remaining three models were based on the feature values calculated from the region 1 kb upstream to TSS plus the whole gene body region, the promoter region (1 kb around the TSS) and the TTS region (± 1 kb) in turn.

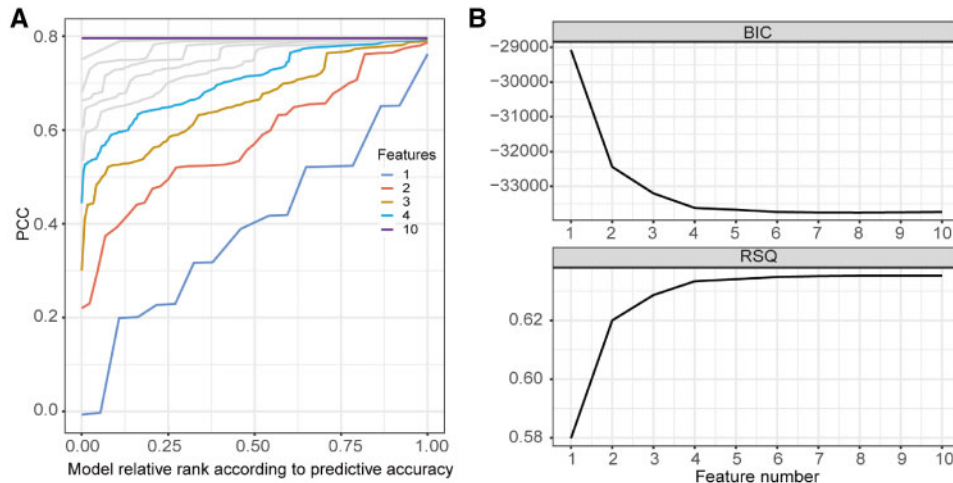


Fig. 2 Predictive power of m -feature(s) linear regression models and feature importance. (A) The predictive power for m -feature(s) models. Models are sorted by ascending predictive accuracy along the x-axis. Lines in gray show the models with feature numbers of five to nine. (B) The relationship between feature number and model complexity measured by BIC or R^2 .

with results obtained using humans (Karlic et al. 2010, Dong et al. 2012), suggesting convergent functions of histone modifications with regard to gene expression regulation in animals and higher plants.

We then examined whether the predictive models perform differently if the feature values were obtained from other genic regions or used in other ways. The results showed that the 'type-aware' scheme outperformed the three alternative schemes (Fig. 1F). We also explored the combinatorial effects of chromatin modifications from different genomic loci toward gene expression levels. To the end, the feature values derived from upstream 1 kb of TSS and gene body were separately calculated, and they were combined as the predictive variables. However, this scheme using twice as many features did not improve gene expression prediction compared with the 'type-aware' scheme (Fig. 1F). So, the 'type-aware' scheme was adopted throughout this study.

To determine whether the gene expression predictions are robust to the input data source, we replaced the original expression dataset with two other RNA-Seq or one microarray data. We found that the predictive model still performed well despite the slight variation in predictive power (Supplementary Fig. S3A). Similarly, when H3K4me3 and H3K27me3 datasets from other sources were used, the prediction power remained high as the original model (Supplementary Fig. S3B, C). Furthermore, the relative importance of each chromatin modification was mostly unchanged (Supplementary Fig. S4). Overall, these observations indicated that the predictive model is independent of data selection.

Chromatin modifications are redundant for predicting gene expression

Several studies have shown that some chromatin modifications are correlated to each other with varying extents (Kouzarides 2007, Zhang et al. 2009, Roudier et al. 2011, Zhang et al. 2015). To determine the extent of the potential redundancy among

these modifications, we built all possible linear regression models by choosing m ($m = 1, 2, 3, \dots, 10$) feature(s) as predictor(s). The performances of the 1,023 models were estimated by 10-fold cross-validation (Supplementary Table S2).

As expected, the predictive accuracy of models increases globally when more modifications were included, but the improvement from the best four-modification model to the full model was limited (Fig. 2A; Supplementary Fig. S5). This pattern was very similar to results obtained in previous studies (Karlic et al. 2010, Roudier et al. 2011, Sequeira-Mendes et al. 2014), which suggested that as few as four chromatin modifications might be sufficient for modeling the expression levels of associated genes in *Arabidopsis*. This finding was confirmed by plotting the number of modifications against the Bayesian information criterion (BIC) or the coefficient of determination (R^2), which revealed that the BIC and R^2 values became steady when using five or more modifications in the model (Fig. 2B). Therefore, our results suggest that a particular combination of four chromatin modifications can be utilized to model the expression levels of associated genes with high accuracy.

Predictive accuracy varies for different genes

Genes with distinctive genomic features may be regulated in distinct manners. First, the predictive performance was compared between using protein-coding and noncoding genes in *Arabidopsis*. The predictive accuracy for noncoding genes was lower than that for protein-coding genes (Fig. 3A). We performed a simulation analysis to assess whether this difference in predictive accuracy was due to the difference of expression noise. Our analysis indicated that the predictive accuracy for the noncoding genes was still significantly below that from the simulated data for protein-coding genes (Supplementary Fig. S6A). Therefore, the expression noise of the noncoding genes cannot explain their relatively poor predictive performance. Besides, to rule out the effect of the sequence length, a simulation test was conducted by sampling protein-coding genes

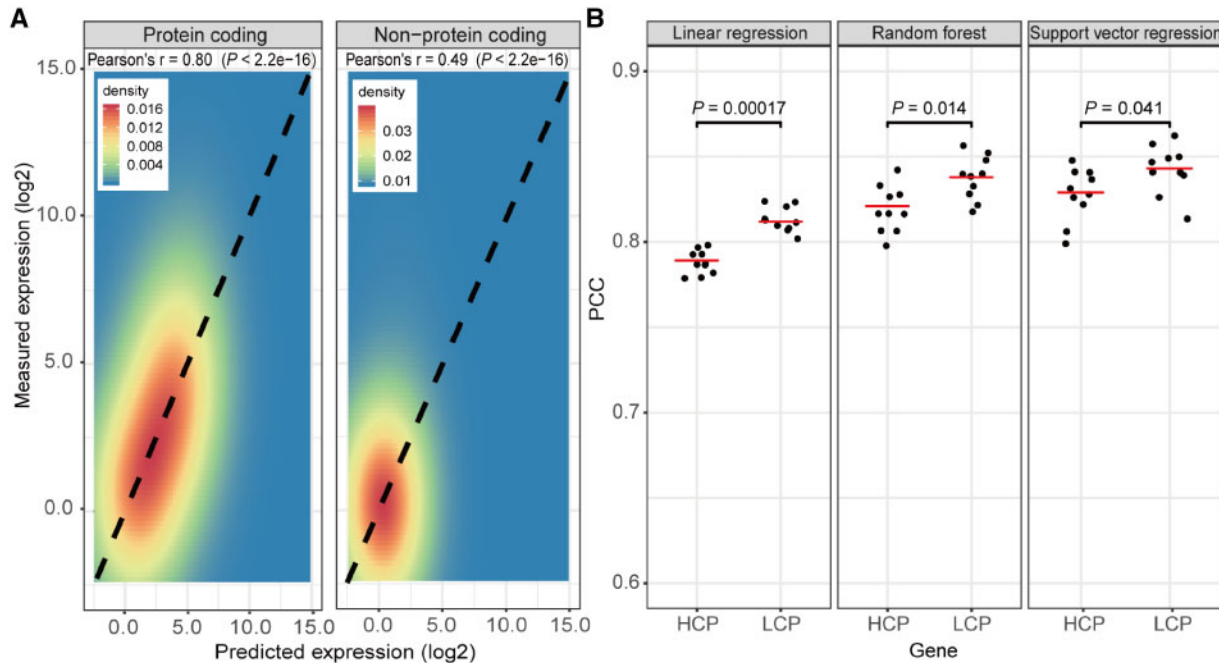


Fig. 3 Comparison of predictive accuracy for different sets of genes. (A) The predictive accuracy for protein-coding and noncoding genes. The black lines show the fit line between the predicted and measured gene expression levels for protein-coding genes (left) and 6,186 noncoding genes (right), respectively. (B) The predictive power for 13,159 HCP and 14,257 LCP genes. The digits show the P -values with respect to the difference in the predictive power between the two gene sets using Wilcoxon rank-sum test (two-sided).

with the same gene length distribution as that of the noncoding genes. Again, the observed predictive accuracy for the noncoding genes was significantly lower than that derived from the simulated data (Supplementary Fig. S6B). Collectively, plant protein-coding genes and noncoding genes may have different transcriptional regulation mechanisms at the level of chromatin modification.

The CpG content in the promoter regions of metazoan protein-coding genes has been recognized as critical to gene function (Lenhard et al. 2012). For example, genes with high-CpG promoters (HCPs) are more likely to be ubiquitously expressed (Chatterjee and Vinson 2012), and HCPs are associated with developmentally regulated genes (Yamashita et al. 2005, Carninci et al. 2006). We observed that the predictive accuracy of HCPs was slightly, but significantly lower than that of low-CpG promoters (LCPs) in *Arabidopsis* (Fig. 3B). Consistently, CpG island-containing genes show a lower predictive accuracy than those genes not associated with CpG island (Supplementary Fig. S7). Therefore, the content of CpG dinucleotide and the CpG island in promoter regions play important roles in regulating gene expression regarding chromatin modifications in *Arabidopsis*.

Predictive performance is highly dependent on the epigenetic features in matched tissues

To test the effect of data matching on the predictive models, we used chromatin modifications in leaf tissue to model gene expression levels in other six tissues: root, endosperm, embryo, flower, silique and shoot. As expected, the predictive models performed best with matched data from leaf tissue (Fig. 4A).

Furthermore, the predictive accuracies on the nondifferentially expressed genes were consistently higher than the differentially expressed genes between leaf tissue and each of other six tissue types (Fig. 4B).

Data for H3K4me2, H3K4me3, H3K9me2 and H3K27me3 were available for root tissue. When we replaced the original levels of the four modifications in leaf tissue with that of root tissue and kept the remaining leaf modification data unchanged, the predictive accuracy for root gene expression was significantly improved (Fig. 4C). Congruently, predictive power decreased when leaf gene expression levels were modeled based on the three root epigenetic marks in addition to the remaining six leaf chromatin features (Fig. 4C). Taken together, these results confirm that gene expression levels in at least seven types of *Arabidopsis* tissue are strongly associated with their levels of chromatin modifications.

We also determined whether the quantitative relationship between gene expression and chromatin modifications is conserved across tissue types. Considering that H3K4me2, H3K4me3, H3K9me2 and H3K27me3 data were available for root and leaf tissue, we used these modifications to build separate linear regression models for leaf and root tissue. Notably, the model trained on leaf data performed well on root data ($r = 0.63$), with predictive accuracy was comparable to that for root tissue by the cross-validation ($r = 0.70$). Conversely, the predictive performance of the model trained on root data for leaf gene expression was nearly equal to that of the model trained using leaf data (Fig. 4D). These results demonstrate that epigenetic modifications play similarly important roles in transcriptional regulation across *Arabidopsis* tissue types.

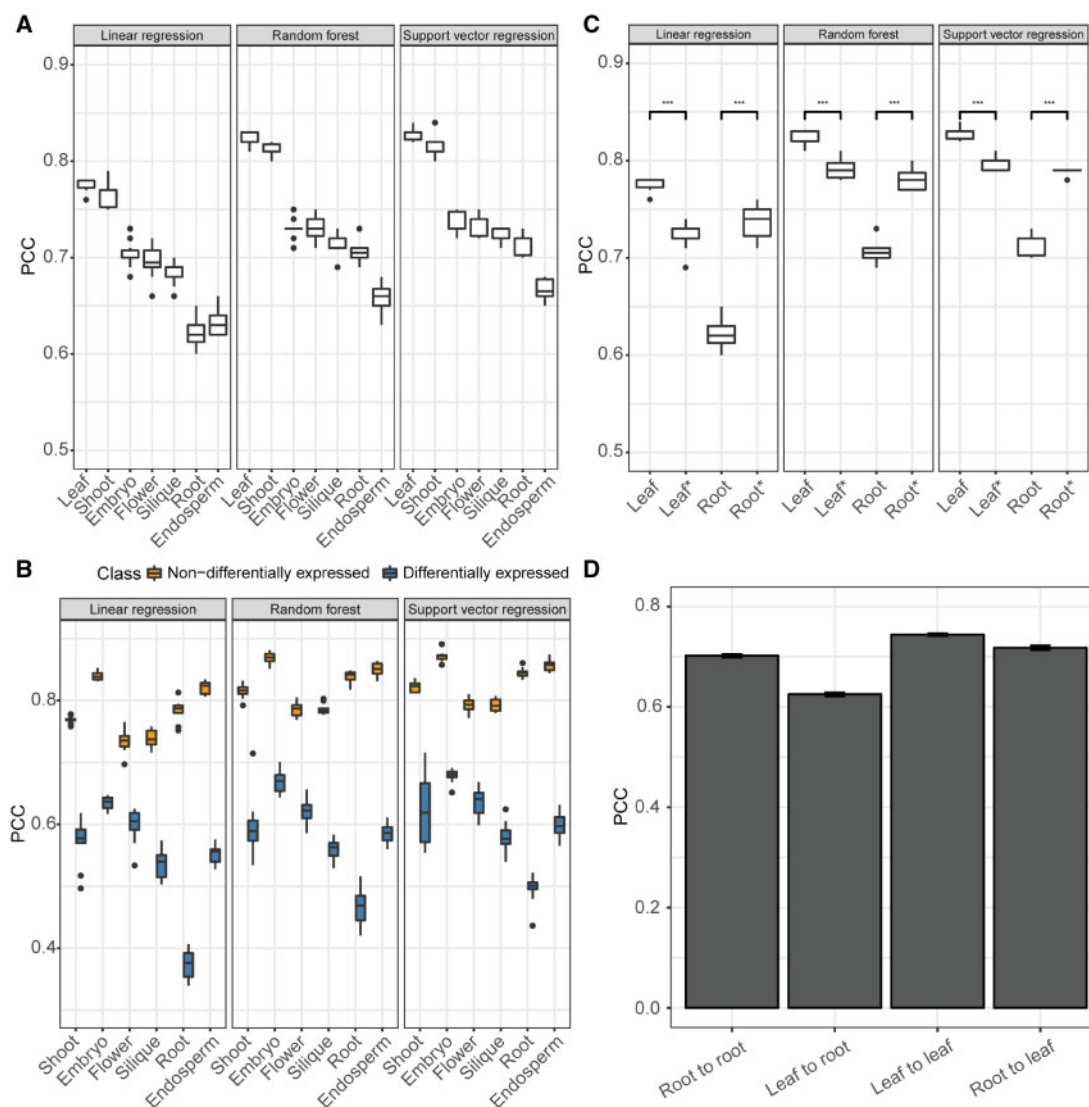


Fig. 4 Effect of data matching on model performance and generalization of the model across tissue types. (A) The model performance for different tissue types using the chromatin modifications of leaf tissue. (B) The predictive accuracy for differentially and nondifferentially expressed genes. For each tissue type except leaf tissue, the genes differentially (blue) or nondifferentially (orange) expressed between this tissue and the leaf tissue were identified. There were 2,668, 15,613, 10,021, 8,970, 10,499 and 16,420 differentially expressed genes identified in shoot, embryo, flower, siliques, root and endosperm tissues, respectively. (C) The predictive power achieved by varying the degree of data matching regarding the expression data and chromatin modification data. *Leaf* and *Leaf** show the predictive accuracy for leaf gene expression using the leaf modification data and its modified version wherein the leaf H3K4me3, H3K9me2 and H3K27me3 levels were replaced with the levels of the same modifications in root, respectively; *Root* and *Root** show the predictive power of gene expression for root tissue using the leaf chromatin modification data and using a mixed dataset: the root H3K4me3, H3K9me2 and H3K27me3 data plus the leaf data of other types of chromatin modifications, respectively. The difference in the predictive accuracy between all pairs of inspected models is statistically significant ($P < 0.001$, two-sided Wilcoxon rank-sum test). (D) The applicability of predictive model across tissue types. Models that were trained using the leaf or root data were used to model the gene expression levels of the same or the other tissue, respectively. The left two groups show the predictive accuracy on the root data for models trained using root and leaf data, respectively; the right two groups show the predictive accuracy on the leaf data for models trained using the leaf and root data, respectively. Data are presented as mean $\pm$ SD.

The effect of distal regulatory elements on gene expression predicting in *Arabidopsis*

Gene transcription is not only regulated by proximal *cis*-regulatory elements promoters, but also by distal regulatory elements such as enhancers. The activity level of enhancers is intimately linked to their epigenetic modifications (Marand

et al. 2017). Thus, we added the chromatin modification data of putative enhancers to the presently used feature list and modeled the gene expression of 4,767 *Arabidopsis* genes that have putative enhancer information available. Compared with the predictive model using chromatin features in the proximal regions of genes, no improvement in the predictive accuracy

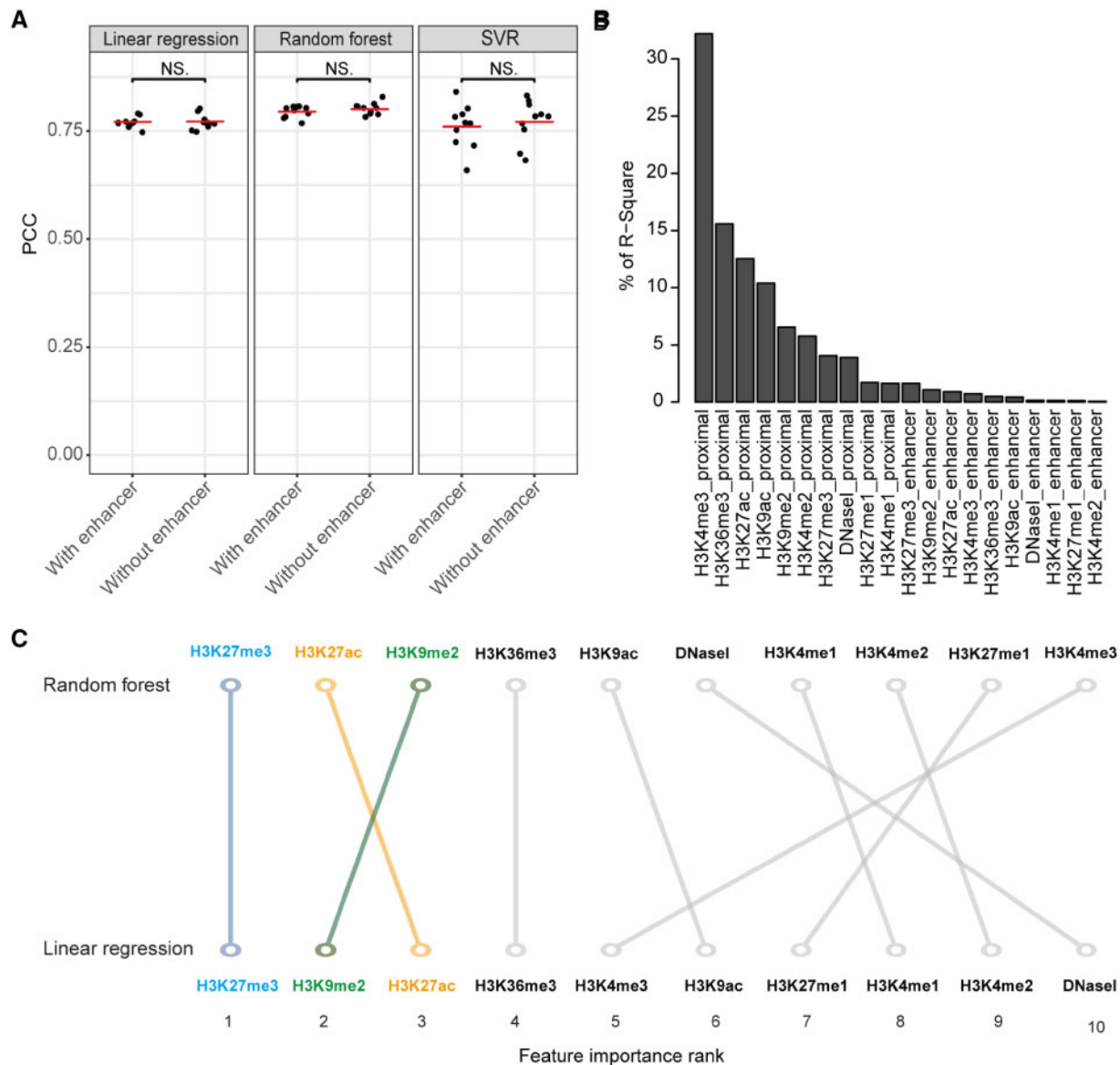


Fig. 5 Effects of putative enhancer on gene expression prediction. (A) The comparison of the predictive accuracy for models with or without using chromatin modifications in enhancers. 'NS' stands for not significant. (B) Feature importance of linear regression model using all chromatin features from the proximal regulatory regions of genes and the putative enhancers. (C) The rank of the feature importance of two models using enhancer information only.

was found (Fig. 5A). Furthermore, the enhancer features seemingly made little and even negligible contribution to gene expression prediction in the integrated model (Fig. 5B; Supplementary Fig. S8), suggesting that proximal regulatory regions have provided enough information about gene expression in *Arabidopsis*. The enhancer information was largely masked during gene expression modeling which could be caused by the fact that the chromatin signals carried by enhancers were similar with those from the proximal regions and were weakly correlated with gene expression. To check this possibility, only the chromatin modification features of the enhancers were used for modeling. Consequently, the average predictive accuracy was 0.34 for linear regression, 0.36 for random forest and 0.37 for SVR. So, to some extent, the enhancer activity can reflect gene expression levels. Remarkably,

H3K27me3, H3K27Kac and H3K9me2 are salient in modeling for both linear regression model and random forest model in *Arabidopsis* (Fig. 5C), suggesting these markers are associated with the intergenic enhancers.

Combination of TFs is predictive of gene expression levels

Our data showed that the DHS intensity within the promoters alone is associated with gene expression level ($r = 0.42$, $P < 2.2 \times 10^{-16}$; Supplementary Table S2). The cooperative binding of TFs in place of canonical nucleosomes results in open chromatin region (Thurman et al. 2012), and the combination of TF binding was predictive of gene expression levels in animals (Cheng and Gerstein 2012, Schmidt et al. 2017). We are curious about

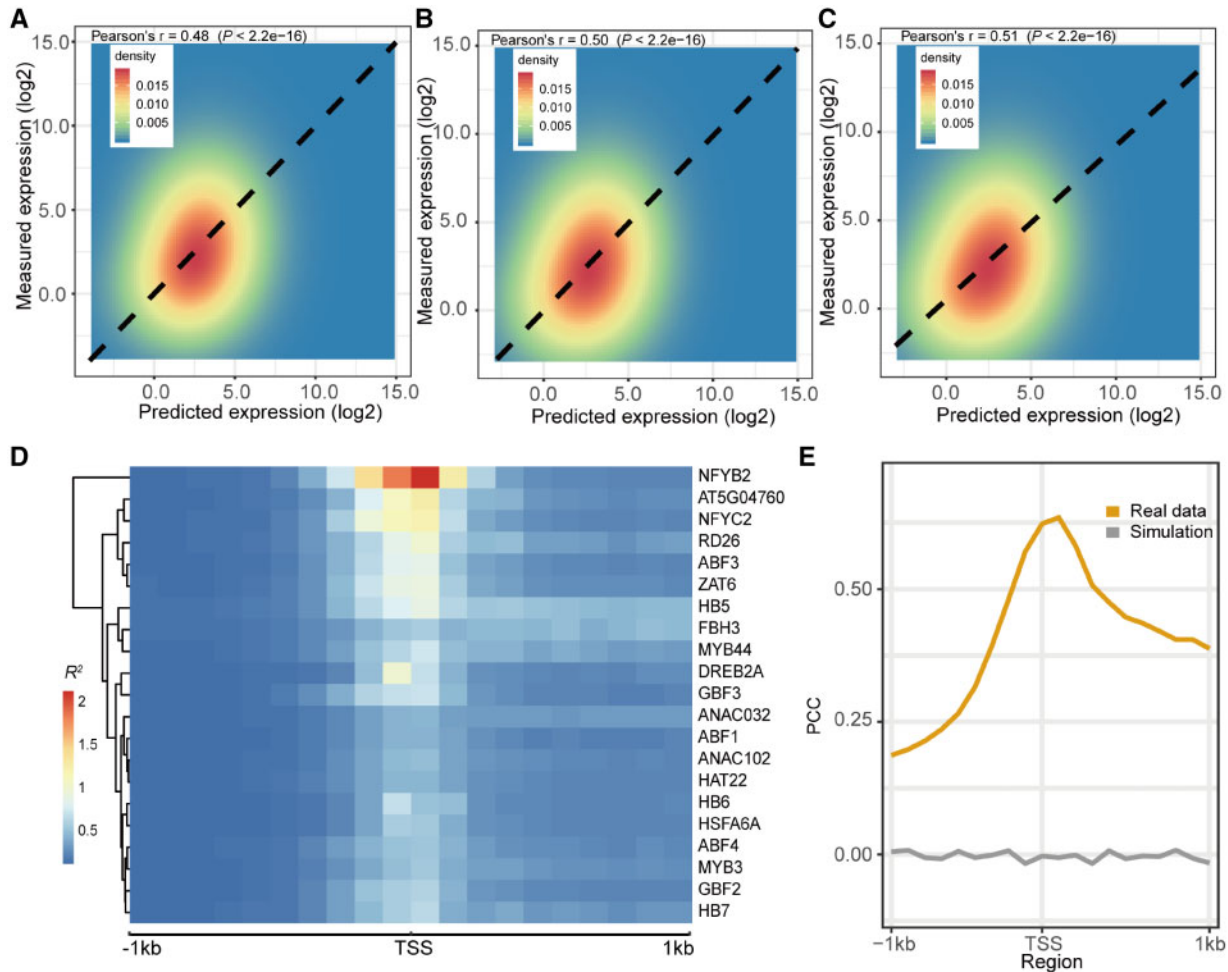


Fig. 6 Predictive accuracy of models on *Arabidopsis* gene expression using TF-binding intensity. (A–C) The predictive accuracies for the linear regression, random forest and SVR model based on 21 TFs-binding intensity, respectively. (D) The heatmap of feature importance measured by R^2 from linear regression model. Each cell represents a genomic bin located in the region of -1 kb to $+1$ kb around the TSS. The normalized R^2 values are shown. (E) The bin-wise predictive accuracy of linear regression model. For each bin, a linear regression model was built to predict gene expression using the binding scores of the 21 TFs in that bin (orange line). By shuffling the corresponding relations between the gene expression data and the TF-binding data, the correlation between TF binding and gene expression disappeared (gray line).

whether a similar rule holds true for plants. Thus, we calculated TF-binding scores based on 21 available ChIP-Seq data of ABA-related TFs and found the predictive models performed well (Fig. 6A–C). Given the relatively limited number of TFs surveyed, it may seem surprising that such a close correlation should be evident. Except two TFs NFYB2 and NFYC2 with a relatively high feature importance, all TFs almost contributed equally to the gene expression prediction (Supplementary Figs. S9, S10). To determine the most informative genomic loci of TF binding, we divided the TSS region (± 1 kb) of protein-coding genes into 20 equal-sized bins and calculated the TF-binding scores for each bin and for each TF. After that, the resultant 21×20 predictors were used to model gene expression levels using linear regression model. Intriguingly, the model performance ($r = 0.69$) got a bit of improvement. The genomic regions between -400 and $+200$ bp contributed a majority of gene expression variance (R^2) in this process (Fig. 6D), which was congruent with the bin-wise predictive performance pattern

(Fig. 6E). Overall, our analysis indicated that the transcription process may be orchestrated by *trans*-regulatory factors in a combined manner where a few TFs have relatively large influence but most TFs are responsible for fine-tuning.

Generalization of the predictive model across plant species

To explore the degree of evolutionary conservation of the relationship between chromatin modifications and gene expression across plants, we analyzed rice data. It revealed that the predictive power in rice was overall lower than that in *Arabidopsis* (Fig. 7A vs. Fig. 1A–C). Nevertheless, consistent with the results in *Arabidopsis*, H3K4me3 and H3K36me3 were still the top two most predictive chromatin marks (Fig. 7B, C vs. Fig. 1D, E), suggesting that the functions of these two histone modifications in gene expression regulation are conserved across plant species.

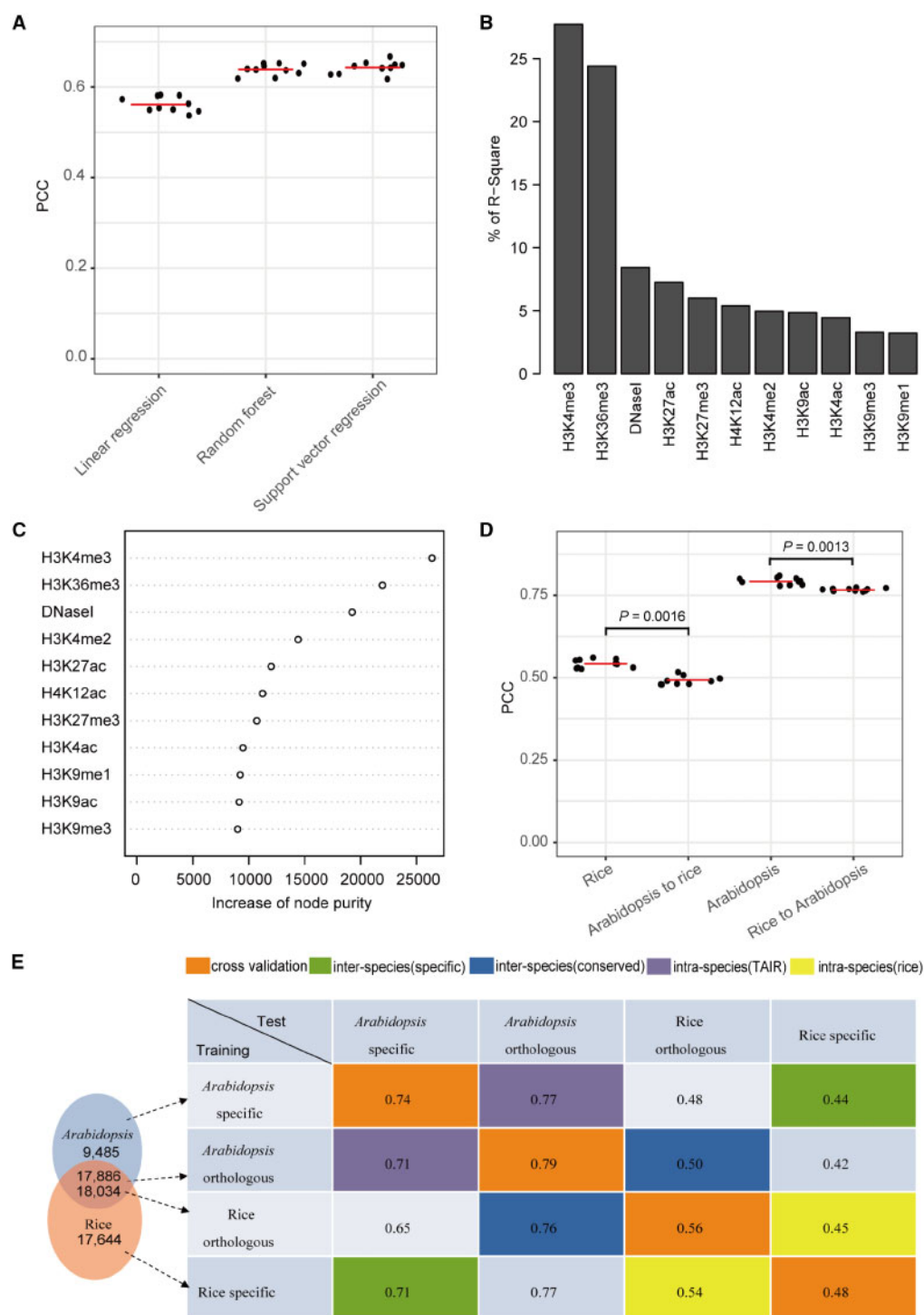


Fig. 7 Prediction of gene expression in rice and between-species comparison of gene expression prediction. (A) Predictive power for models using rice chromatin modifications and gene expression data. Dots in black indicate the 10 predictive power values by cross-validation process. The lines in red denote the mean values. (B, C) The feature importance of the linear regression model and the random forest model, respectively. (D) Generalization of predictive model for gene expression across species. Groups 1 and 2 show the predictive accuracy on the rice data for models trained using the conspecific and the allospecific data, respectively; Groups 3 and 4 show the predictive power on the *Arabidopsis* data for models trained using the conspecific and the allospecific data, respectively. (E) The Venn diagram shows the numbers of genes that are either species-specific or orthologous to genes in the other species. The table shows the predictive power obtained for each gene set according to the models trained on the data from the same or the other gene sets. The data in box were used for demonstration in the main text. Obviously, for the *Arabidopsis* orthologous genes, the predictive accuracy ($r = 0.79$) of their expression levels calculated by cross-validation process were nearly equal to that achieved by the conspecific model ($r = 0.77$ using *Arabidopsis*-specific genes as training set) and that obtained by the two allospecific models ($r = 0.76$ and $r = 0.77$ using rice orthologous genes and rice-specific genes as training set, specifically).

Next, we examined the applicability of a model trained in one species to another species. Due to the good interpretability, the linear regression model was used to probe the relationship between gene expression and the seven chromatin modifications that were available for both *Arabidopsis* and rice. For each species, two models were used to model the quantitative relationship between gene expression and chromatin modification: one was trained using conspecific data and the other was trained using allospecific data. As expected, the predictive power of the allospecific model was consistently lower than that of the corresponding conspecific model, but the difference was relatively small for both species (Fig. 7D). Therefore, given the chromatin modification data of one plant species, it seems feasible to predict the corresponding gene expression levels with a model trained in another plant species with a limited reduction in accuracy.

To understand the effects of gene orthology on model power, we categorized all protein-coding genes of both species into four groups: *Arabidopsis*-specific, rice-specific, inter-species conserved rice or inter-species conserved *Arabidopsis* genes. We separately trained linear regression models using gene expression data and chromatin modification data pertaining to each gene set, after which the model performance was estimated for the same set and other gene sets. In general, the association between gene expression and chromatin modifications in *Arabidopsis* seemed to be stronger than that of rice (Fig. 7E; Supplementary Fig. S11). Interestingly, conserved genes were seemingly more easily to fit the predictive model than species-specific genes (cross-validation r : 0.56 vs. 0.48 in rice and 0.79 vs. 0.74 in *Arabidopsis*). We noted that the number of species-specific genes was less than that of the orthologous genes in each species. To eliminate the influence of this sort of unbalance, we sampled the orthologous genes the same number as the set of species-specific genes and recalculated the predictive accuracy, the similar result was obtained (Supplementary Fig. S12).

Moreover, the expression levels of genes in each gene set were well fitted by the model trained using another conspecific gene set, as well as by the other two models trained using allospecific data, both of which had performance comparable to the cross-validation. For example, the predictive accuracy estimated by cross-validation for the *Arabidopsis* orthologous genes was nearly equal to the predictive accuracy of the other three models trained using different gene sets (Fig. 7E, the column boxed). Overall, our analysis demonstrates that a conserved molecular mechanism may be responsible for the close relationship between gene expression and chromatin modifications across the two species.

Signal profiles of chromatin modifications mirror their feature importance

To corroborate the observed relative importance of chromatin features in gene expression modeling, we divided all *Arabidopsis* protein-coding genes into three gene sets with low, medium or high expression levels and compared the three signal profiles of each chromatin modification using leaf dataset (Fig. 8A).

Notably, the more important of a chromatin modification in gene expression prediction, the greater disparity was seen among its signal profiles with respect to the three gene sets. For example, for active chromatin modifications H3K4me3 and H3K36me3, their respective three profiles varied greatly, with the relative peak height positively related to gene expression level.

Notably, most profiles of histone modifications of the same type are highly resembling between *Arabidopsis* and rice (Fig. 8). Meanwhile, the DHSs are predominantly positioned immediately upstream to TSS and this spatial effect is independent of mRNA abundance (Fig. 8). Therefore, our study suggests that the establishment of a conserved epigenomic mechanism underpinning the transcriptional control of gene expression preceded the divergence of the monocots and dicots approximately 40–150 million years (Myr) ago (Chaw et al. 2004).

Discussion

Gene expression is tightly controlled at multiple levels. Deciphering the histone code to the transcription process is fundamental to our understanding of the gene expression regulation. There have been intensive studies on the quantitative relationship between gene expression and up to tens of different chromatin modifications in animals (Karlic et al. 2010, Cheng et al. 2011, Cheng and Gerstein 2012, Dong and Weng 2013, Singh et al. 2016), but similar studies in plants only focused on limited types of chromatin modifications (Zhang et al. 2009, Zong et al. 2013, Zhao et al. 2018). To our knowledge, this is the first systematic and integrative analysis to elucidate the complex relationship between gene expression and over 10 types of chromatin modifications in plants by using quantitative modeling.

The primary challenge of this study is to determine the regions and the manner to quantify chromatin modification levels for each gene that convey biological significance. However, in this respect, existing studies that mostly focused on animals have not developed a unified method (Karlic et al. 2010, Costa et al. 2011, Cheng and Gerstein 2012, Dong et al. 2012, Schmidt et al. 2017). A detailed discussion was reviewed recently (Huminiecki and Horbańczuk 2017). By plotting the signal profiles of the chromatin modifications around the genic region of *Arabidopsis* genes in leaf tissue (Supplementary Fig. S1), we got well known about the spatial mode of these signals, which lay a foundation for measuring the chromatin modification levels of genes. Accordingly, the ‘type-aware’ scheme formulated by us produced the same best model performance as the ‘Upstream 1 kb + Gene body’ scheme that took the combinatorial effects of two genomic loci toward expression levels into account. So, this result implies either that these chromatin modifications are preferentially deposited at particular genomic region or that the epigenomic information conveyed in this particular way is highly redundant toward gene expression levels. We reckon that the former should be much more reasonable as evidenced by the signal profiles of chromatin

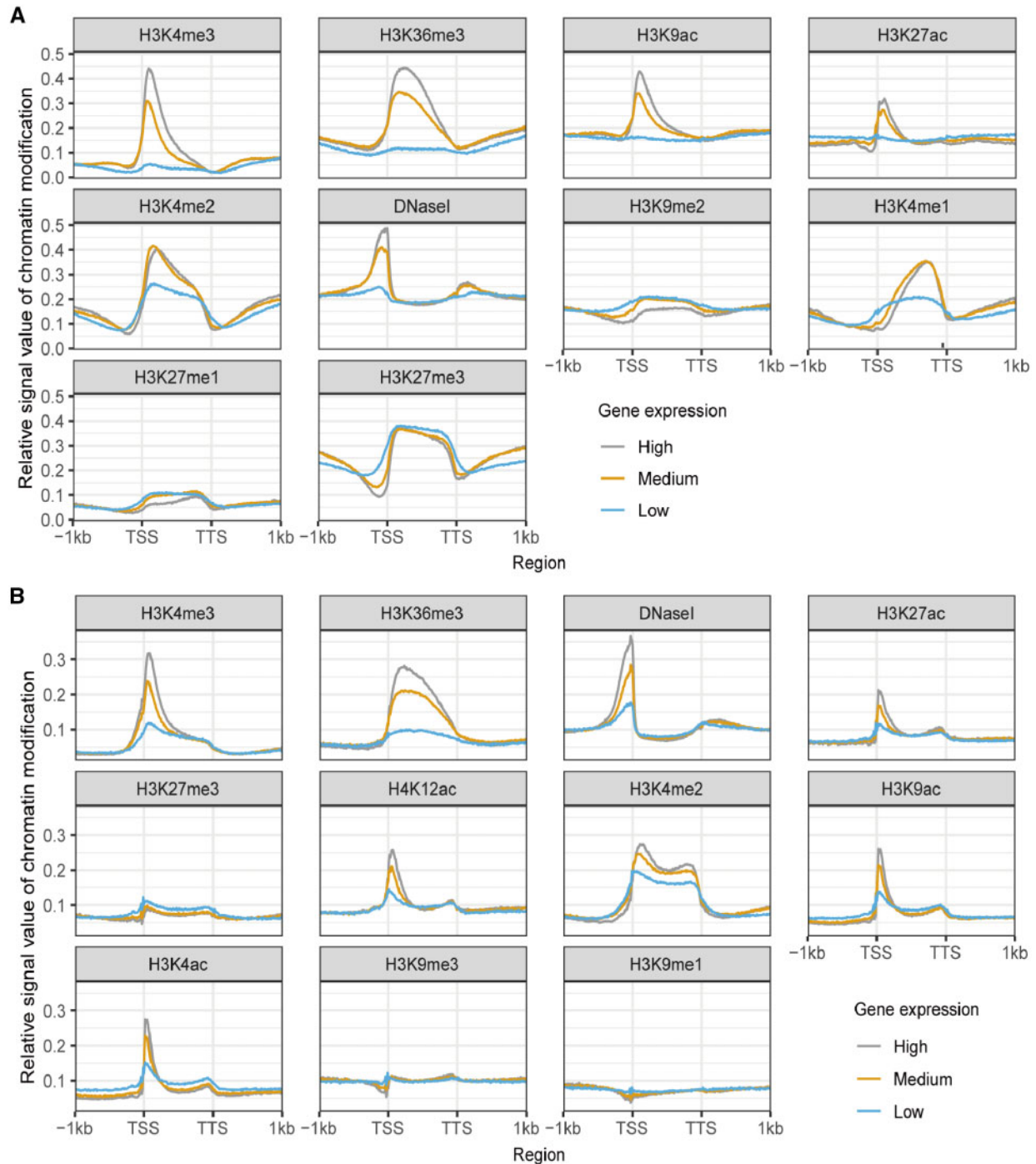


Fig. 8 Comparison of signal profiles for each chromatin modification for protein-coding genes with low, medium or high expression. (A, B) The chromatin modification profiles in *Arabidopsis* and rice, respectively. In brief, gene body regions and their 1 kb upstream and 1 kb downstream regions were separately divided into 100 equal-sized bins for each gene. Then, the read counts in each bin were counted and normalized by the bin size. To obtain the relative signal intensity of each chromatin modification in each bin, the normalized read number for each bin was further divided by the maximum read count for each gene. Finally, the signal densities were averaged over the protein-coding genes with low, medium and high expression levels, respectively.

modifications (Fig. 8). In addition, we have found that using either H3K4me3 or H3K36me3 alone achieved relatively high predictive power (Supplementary Table S2). H3K4me3 and H3K36me3 are usually the marks of active genes and mainly involved in transcriptional activation and elongation, respectively

(Lawrence et al. 2016). However, using H3K4me3 and H3K36me3 together did not improve model performance ($r = 0.76$) relative to that achieved using H3K4me3 only ($r = 0.76$), suggesting that these chromatin modifications harbor highly redundant information about gene expression levels.

Our data showed that the protein-coding genes were more predictable than noncoding genes (Fig. 3A). This could be caused by different transcription machinery: the transcription of protein-coding genes is initiated by RNA polymerase II, while most noncoding genes are transcribed by RNA polymerase I and III. The expression levels of *Arabidopsis* LCPs were easier to predict than the HCPs, which is opposite to the results in human cells (Dong *et al.* 2012). We speculate that this disparity may be due to the difference in promoter architecture between *Arabidopsis* and human: the former displays a bimodal distribution with regard to the promoter CpG content for all protein-coding genes (Saxonov *et al.* 2006), but the latter exhibits an approximately normal distribution (Supplementary Fig. S13).

We revealed a high level of transitivity in the relationship between gene expression and the chromatin modifications across *Arabidopsis* tissue types. Also, the quantitative relationships between chromatin modifications and the expression are conserved across different plants (Fig. 7D). Concordantly, the signal profiles of the two most predictive modifications, H3K4me3 and H3K36me3, were highly similar between *Arabidopsis* and rice (Fig. 8). More interestingly, we discovered that the expression levels of orthologous genes were easier to predict than those of species-specific genes (Fig. 7E). Thus, multiple lines of evidence have convinced us that the quantitative relationship between gene expression and chromatin modifications was established prior to the divergence between monocots and dicots and concomitantly shaped the evolution of gene expression.

Our results showed that the overall model performance in rice was inferior to that in *Arabidopsis*. One possible reason is the fact that the sequencing depth for the same type of chromatin modification data in rice is generally lower than that in *Arabidopsis* (Supplementary Table S3). By scaling all the data to 2× sequencing depth, the predictive models performed comparably as the models using the original data for both species (Supplementary Fig. S14). Therefore, the difference in the sequencing depth cannot explain the difference in predictive accuracy of gene expression between the two species. The second possible reason is the matching degree between the expression and chromatin modifications data. The present integrative analysis was based on multiple chromatin modification datasets from different laboratories. Thus, the heterogeneous data may reduce the degree of data matching for the tissue samples used for RNA-Seq and chromatin modification profiling in our study. Additionally, a previous study revealed that the regulatory complexity is positively related to the amount of noncoding DNA flanking the genes (Nelson *et al.* 2004). Accordingly, the longer and higher proportion of intergenic regions in the rice genome than that of *Arabidopsis* (Supplementary Fig. S15) may imply a much more complex gene expression regulation in rice. Consistent with this hypothesis, the Hi-C data analysis showed that there is no clear TAD (Topologically Associating Domain) structure in *Arabidopsis* (Feng *et al.* 2014), whereas thousands of distinct TADs account for about a quarter of the rice genome (Liu *et al.* 2017). Within TADs, frequent chromatin interactions bring distal regulatory elements like enhancer or insulator into the promoter, leading

to an improved precision and robustness of gene transcription and expression. Therefore, we speculate that the difference in gene regulation complexity between the two plants may partially account for the difference in predictive accuracy. But this problem could be solved only if more high-quality distal regulatory elements are identified in plants.

Gene transcription is not only regulated by proximal *cis*-regulatory elements promoters, but also by distal regulatory elements such as enhancers. Though the study of plant enhancers has long been lagged behind that of animal systems, it has been highly appreciated recently that the identification and characterization of plant enhancers will become a promising and fundamental field in plant biology in the coming decades (Weber *et al.* 2016, Marand *et al.* 2017). Studies using animals have revealed that the multiple enhancers targeting a common gene can encode redundant regulatory information toward each other to ensure the precision of gene expression pattern (Perry *et al.* 2010, Perry *et al.* 2011). Interestingly, in this study, the chromatin features in enhancers alone were moderately associated with expression levels of *Arabidopsis* protein-coding genes, but at the global level, they cannot improve the predictive accuracy of the original models that used the features from proximal regions only. Therefore, besides the known function of enhancers in directing tissue-specific gene expression pattern (Shen *et al.* 2012), our results may shed light on a new layer of regulatory complexity: the interaction between enhancer and promoter is likely to confer target genes a reliable expression abundance in response to environmental or genetic perturbations.

In this study, we also investigated the possibility of predicting target gene expression levels by TF-binding data in *Arabidopsis*. The observed high correlation between TF-binding score and gene expression level is striking given that the model features were derived from only 21 TFs, which is reminiscent of the observation that the accumulative ChIP-Seq signal from only 42 TFs parallels the DHS signal in human cells (Thurman *et al.* 2012). Furthermore, this predictive accuracy just gave a lower bound due to the imperfect tissue match between the ChIP-Seq data and the expression data where the former were obtained from the whole seedling of *Arabidopsis* but the latter were from leaf tissue type. We chose to use the ChIP-Seq data at the cost of tissue matching because of the limited availability of measurements for a batch of TFs in one experiment. Consistent with an early finding that *Arabidopsis* TF-binding sites were densely located in the genomic region −400 to +200 bp to TSS (Yu *et al.* 2016), our analysis showed that the TF-binding signals within this region were strong predictors of gene expression level, providing evidence that the combinatory effect of TFs is highly predictive of gene expression levels in plants.

Overall, our analysis and other relevant studies have shown that gene expression levels are reflected a large extent by levels of chromatin modifications in angiosperms and animals. However, the molecular mechanisms linking chromatin modifications and gene regulation are poorly understood. In a recent review, Lawrence *et al.* (2016) proposed that histone modifications can be drivers of transcriptional change with causative functions rather than mere by-products of transcription or

marks of the chromatin state. It is well-documented that the expression dynamics of genes is governed by the interaction between *cis*-regulatory elements and *trans*-factors as well as nucleosome (Segal and Widom 2009). Epigenetic modification of chromatin can remold the TF affinities with *cis*-regulatory elements, therefore, it adds the plasticity of gene expression for cells to respond to physiological and environmental cues and influences phenotype. Actually, TF-binding sites have been utilized for modeling gene expression levels and the epigenetic state of cells (Whitaker et al. 2015, Schmidt et al. 2017). However, more work is needed to build a comprehensive understanding of the mechanisms underlying gene regulation and decipher the relationship between gene transcription and epigenetic modifications, especially in plant species.

In conclusion, this work systematically investigated the combinatorial effects of multiple epigenomic features on gene expression in *Arabidopsis* and rice using machine-learning-based methods. Although this concept is not novel, we took a new angle on studying the tissue- and species-wide conservation of the quantitative association between chromatin modifications and gene expression, whereby revealing a shared epigenetic mechanism underlying gene expression regulation. We also determined the minimal number of chromatin modifications types that together could accurately predict the gene expression abundance. Additionally, the better predictive performance in *Arabidopsis* than in rice may reflect their difference in gene regulatory complexity. More importantly, whether the observed quantitative relationship between the epigenomic modifications and gene expression is conserved or not in other plants can be easily evaluated using the current machine-learning-based framework only if sufficient data of chromatin modifications and gene expression for the species of interest can be collected.

Materials and Methods

Arabidopsis and rice data collection and processing

Arabidopsis reference genome (TAIR10) was downloaded from the Ensembl Plant database (<http://plants.ensembl.org/>, April 5, 2018, date last accessed). ChIP-Seq data for histone modifications H3K4me1, H3K4me2, H3K4me3, H3K9me2, H3K9ac, H3K27me1, H3K27me3, H3K27ac and H3K36me3 were collected from recent publications (Zhang et al. 2012b, Chica et al. 2013, Brusslan et al. 2015, Moreno-Romero et al. 2016, Zhang et al. 2016, Chen et al. 2017, Engelhorn et al. 2017, Sanders et al. 2017). DNase-Seq data for DHS was obtained from a previous study (Zhang et al. 2012b). All these data were profiled using *Arabidopsis* leaf tissue and the detailed information about these data is included in Supplementary Table S1.

Raw reads were filtered by read quality, after which adapter sequences were trimmed if detectable. Clean reads for each chromatin modification were separately mapped to TAIR10 using Bowtie2 with defaults (Langmead and Salzberg 2012). Then, the uniquely mapped reads were kept and the duplicated reads were filtered. To calculate the chromatin modification level for each gene, the mapped reads of certain chromatin modification within a defined region were summed, after which this count was normalized by the total number of mapped reads and the length of the region as the signal intensity (FPKM). We separately calculated the chromatin modification intensity within four genomic regions: the promoter region (1 kb around the TSS), 1 kb upstream of TSS, gene body [from TSS to transcription termination site (TTS)] and the TTS region (1 kb around the TTS).

The leaf transcriptome data were profiled using RNA-Seq or microarray techniques (Supplementary Table S1). For RNA-Seq data, the raw reads were preprocessed as described above, after which the resulting clean reads were aligned to the reference genome with TopHat2, with two mismatches allowed (Trapnell et al. 2009). The alignment results were processed using Cufflinks (Trapnell et al. 2012) for gene expression quantification (FPKM). Alternatively, read-count-based gene expression levels were obtained for each gene using HTseq (Anders et al. 2015). A set of processed microarray data from *Arabidopsis* leaf tissue (accession number AT40) was also downloaded from the Plant Expression Database (Dash et al. 2012).

For comparison, rice chromatin modifications and RNA-Seq data in leaf tissue were collected (Supplementary Table S1). In brief, the RNA-Seq and H3K4me3 data were obtained from SRA database (PRJNA391551) (Dong et al. 2017). The ChIP-Seq data for H3K4ac, H3K9me1, H3K9me3, H3K9ac, H3K27ac and H3K27me3 were obtained from SRA database (SRP071339) (Fang et al. 2016). H3K4me2, H3K36me3, H4K12ac and DNase-Seq data were obtained from GEO database (GSE26610) (Zhang et al. 2012a). Particularly, H3K4me3, H3K36me3, H3K27ac, H3K9ac, DNase I, H3K4me2 and H3K27me3 datasets were available in both *Arabidopsis* and rice datasets. The rice data were processed in the same way utilized for the *Arabidopsis* data except that the reads were mapped to rice reference genome MSU-7.

Arabidopsis enhancer data collection

The loci information of putative leaf enhancers in *Arabidopsis* were obtained from Zhu et al. (2015). These enhancers were predicted as the DHSs located in intergenic regions (downstream of transcription terminal site and >1.5 kb upstream of TSS) and their putative target genes were assigned in the original study. Totally, we compiled 7,515 leaf enhancers which are associated with 4,767 target genes. If multiple enhancers are associated with a target gene, we picked the nearest one and calculated its chromatin modification levels using the same method as described for the proximal regions of genes and the values were corresponded to the target gene.

Estimation of TF-binding score from ChIP-Seq data

The raw ChIP-Seq data of 21 *Arabidopsis* TFs (AT1G22640, AT1G49720, AT1G51140, AT1G56170, AT1G77450, AT2G22430, AT2G46270, AT2G46680, AT3G19290, AT4G01120, AT4G27410, AT4G34000, AT4G37790, AT5G04340, AT5G04760, AT5G05410, AT5G43840, AT5G47640, AT5G63790, AT5G65310 and AT5G67300) were obtained from a previous study (Song et al. 2016). This dataset provides whole genome-wide TFs-binding patterns in whole seedlings. The processing of these data is the same as the analysis of histone modification data. The overall binding score of each TF for a gene was calculated by counting the number of mapped reads fallen in the region 1 kb upstream to the TSS of this gene and divided by the total mapped reads number.

Classification of protein-coding genes

We referred to a previous study to classify 27,416 annotated *Arabidopsis* protein-coding genes into HCP or LCP genes (Saxonov et al. 2006). Genes with normalized CpG content in promoter regions (1 kb around the TSS) greater than the average value were classified as HCP, while the remaining genes were classified as LCP. Totally, 13,159 HCP and 14,257 LCP genes were obtained.

Arabidopsis protein-coding genes were also classified based on the presence or absence of CpG islands in their promoter regions. First, we identified all CpG islands in the *Arabidopsis* genome using CpGScan (Jian et al. 2017) by the following criteria: GC content $\geq 55\%$, ObsCpG/ExpCpG ≥ 0.60 , and length ≥ 200 bp. Protein-coding genes that overlapped with any CpG islands were classified as CpG island-positive genes, whereas genes that did not overlap with CpG islands were classified as CpG island-negative genes. The classification process yielded 12,800 CpG island-positive genes and 14,616 CpG island-negative genes.

Identification of differentially expressed genes between leaf and other tissues

The root, shoot, inflorescence and silique RNA-seq data were collected from Liu et al. (2016) (SRP047300), and the endosperm and embryo data were from Hsieh et al. (2011) (SRP003799). The data were processed in the same way as

the leaf data to obtain read-count-based gene expression levels. Then, raw read-count matrix was normalized using the weighted trimmed mean of *M*-values (TMM) method implemented in edgeR (McCarthy et al. 2012), after which differentially expressed genes showing at least 2-fold differences ($FDR < 0.05$) between leaf and other tissues were identified.

Cross-species orthologous genes and species-specific gene

Orthologous gene information for rice and *Arabidopsis* were retrieved from the Biomart database (Aken et al. 2016). Then, four sets of genes were produced: (i) 9,485 *Arabidopsis*-specific genes, (ii) 17,886 *Arabidopsis* genes that are orthologous to rice genes, (iii) 17,644 rice-specific genes and (iv) 18,034 rice genes that are orthologous to *Arabidopsis* genes.

Model building for gene expression prediction

The signal intensity of chromatin modification *j* for gene *i* was X_{ij} . For all chromatin modifications, transformed X_{ij} values [$\log_2(X_{ij}+1)$] were used as predictor variables to train models and predict the logarithm of each gene expression level. Three machine-learning models were used: multiple linear regression, random forest and SVR. The performance of each model was evaluated by 10-fold cross-validation. The Pearson correlation coefficient *r* between predicted and measured gene expression values was calculated. The predictive power or accuracy of a model was calculated as the average of the 10 *r* values.

The *lm* function implemented in R was used to perform linear regression analysis. Feature importance of linear regression model was determined by measuring the relative contribution to R^2 . We also utilized random forest and SVR models as implemented in the R packages randomForest and e1071, respectively (Meyer 2013, Liaw and Wiener 2015). By plotting the error rate for the random forest model with a series of tree numbers, 200 trees ($n \text{ tree} = 200$) that made the prediction error rate just steady were subsequently used in modeling. The feature importance was weighted by the total decrease in node impurities measured by the residual sum of squares, which was computed using the importance function of the randomForest package. For the SVR model, the nonlinear radial basis kernel was used to train the predictive model. The scripts for this work are available on GitHub (<https://github.com/Zefeng2018/Plant-Gene-Expression-Prediction>).

Supplementary Data

Supplementary data are available at PCP online.

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Disclosures

The authors have no conflicts of interest to declare.

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